Beşir ÖZDEMİR

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Summary

Mechanical Engineering student at Yıldız Technical University specializing in structural analysis, FEA, and airframe design. Founded Samsun's first UAV team and led structural design for a Teknofest-finalist rocket team, delivering flight-ready hardware under competitive conditions. Proficient in ANSYS, CATIA, SolidWorks, XFLR5, and OpenRocket.

Education

Yıldız Technical University İstanbul
B.Sc. Mechanical Engineering. 08/2024 - Present
Relevant Coursework: Structural Analysis · Fluid Mechanics & Aerodynamics · Manufacturing Methods · Engineering Mathematics

DENEYAP Makers Lab Samsun
The first cohort of the Samsun DENEYAP Workshop 05/2021 - 07/2024
Completed a 3-year, 11-discipline technology program that developed practical engineering project skills across robotics, electronics, and software domains.

Atakum Imam Hatip High School Samsun
GPA: 91.74 / 100 09/2018 - 07/2023

Experience

Eyüpsultan DENEYAP Makers Lab İstanbul
Teacher 10/2026 – Present

- Delivering hands-on training in design and manufacturing, introducing students to fundamental engineering principles and practical production techniques.
- Guiding students through design-oriented activities and practical projects, fostering technical problem-solving, creativity, and engineering thinking.
- Completing a structured five-stage instructor development and qualification process to ensure readiness for effective technical instruction.
- Preparing to expand instructional responsibilities across 11 additional training modules, broadening technical expertise across multidisciplinary technology fields.

Sampa Otomotiv Samsun
Mechanical Engineering Intern 07/2026 – 08/2026

- Observed and analyzed manufacturing processes for heavy-vehicle components, including hot forging, machining, and bushing assembly.
- Developed an integrated preliminary cost estimate and manufacturing process flowchart for a hydraulic cylinder component, using CAD models, dimensional measurements, material data, and production requirements to support cost analysis and process planning.
- Investigated the design and testing requirements of heavy-vehicle hub cover components, including fixture concepts for pneumatic leak testing.
- Designed a foldable packaging workstation in CAD, incorporating dimensional measurements, material availability, and ergonomic requirements.

Projects

TÜBİTAK High School UAV Competition Samsun
CAD & Structural Design Lead 02/2020 – 09/2020

- Secured national competition qualification for Samsun's first-ever UAV team by leading all structural design and technical documentation, achieving a Design Report score of 92/100 — top 15 & among +100 competing teams.
- Delivered flight-ready CAD models and manufacturing drawings under pandemic constraints, enabling the team to earn field-competition rights and formal recognition from the judging panel.
- Led mechanical development as the sole specialist in a 6-person multidisciplinary team, successfully executing the transition from CAD to fabrication and finalizing flight-ready documentation under a tight 7-month schedule.

Teknofest High School Rocket Competition Samsun
Team Captain & Structural Design Lead 10/2021 - 08/2022

- Led the first rocket team established within Samsun DENEYAP to reach the national final, competing against 100+ teams by owning end-to-end structural integrity analysis, CAD modelling, and manufacturing process execution.
- Produced all technical drawings and GD&T-referenced; documentation for rocket airframe components, ensuring structural compliance through iterative design reviews prior to fabrication.
- Directed 3-discipline team (propulsion, structures, avionics) through full design-to-launch cycle; rocket reached national final among 100+ competing teams.
- Performed structural integrity analysis of rocket airframe in Solidworks Simulation; iterated wall thickness design based on FEA results prior to fabrication.
- Conducted trajectory and stability simulations in OpenRocket to validate airframe design prior to structural review

Skills & Interests

Tools: SolidWorks (CSWA Certified) · CATIA V5 (Intermediate) · ANSYS Mechanical (Intermediate) · XFLR5 (Intermediate) · MATLAB (Intermediate) · OpenRocket (Intermediate) · Python (Intermediate) · Microsoft Office
Technical: FEA / Structural Analysis · Aerodynamic Analysis · UAV Structural Design · Technical Drawing / GD&T · Manufacturing Methods · Process Automation (Python)
Language: Turkish (native) · English (B2)
Interests: Violin · Psychology · Neuroscience